Assignment No.: 03

Q.1) What is Node.js?

→ Node.js is a javascript runtime environment that allows developers to execute Javascript code outside the browser. it is built on Chrome's V8 engine and is mainly used for building server side applications.

key Features:

- Asynchronous & Non-blocking: Handles multiple requests efficiently.
- single-threaded: User event driven architecture
- fast Execution: Powered by the V8 engine.
- Cross-platform: Runs on Windows, macOS, & Linux.

Q.2) What is Express API?

→ Express.js is a minimal and flexible web framework for Node.js used to build web applications and APIs

Express APIs refers to an API built using Express.js to handle HTTP request like GET, POST, PUT & DELETE

Why Use Express.js?

- Simplifies Routing: Handles multiple request types
- Middleware support: Extends functionalities.
- Fast Development: Lightweight and efficient.

Example of a basic Express API:

```
const express = require("Express");
const app = express();

app.get('/', (req, res) => {
    res.send('Hello, world!');
});
app.listen(3000, () => console.log('server running on
                                     port 3000'));
```

Q.3) What is MongoDB?

→ MongoDB is a NoSQL database that stores data in JSON like documents insted of tables. It is highly scalable and used in model applications.

key Features.

- Flexible Schema: No fixed table structure
- scalability: Handle large datasets efficiently
- High performance: Fast read/write operation.

MongoDB store data in Collections and documents

Example of a MongoDB Collection's document:

```
{
    "name": "John Doe",
    "email": "john@gmail.com",
    "age": 20
}
```

Q.4) What is CRUD Operation? Explain in Detail

→ CRUD stands for Create, Read, Update and Delete - the Four basic operations performed on a database.

1. Create (Insert Data)
Used to add new data to a database.
MongoDB Example:
db.users.insertOne({name: "Alice", age: 25});

2. Read (Retrieve Data)
Used to fetch existing data.
MongoDB Example
db.users.find({name: "Alice"});

3. Update (Modify Data)
Used to Modify existing records.
MongoDB Example:
db.users.updateOne({name: "Alice"}, {$set: {age: 26}});

4. Delete (Remove Data)
Used to delete records from the database
db.users.deleteOne({name: "Alice"});